

INFORMS 2026 Data Mining Society Data Challenge

Weather Variables: Sources, Units, and Aggregation Methods

Hybrid Weather Source

Weather features are drawn from two complementary sources to balance spatial resolution and variable breadth:

Source	Description
NOAA URMA (2.5 km)	Wind and temperature. The Unrestricted Mesoscale Analysis is an observation-corrected analysis at 2.5 km resolution. Used for the high-value wind and temperature fields that drive outages. Extracted as polygon mean over each county (uses full 2.5 km resolution, not a single point). This corrects the wind-gust inflation artifact seen in coarser reanalysis.
ERA5 via Open-Meteo (~31 km)	Supplementary fields URMA does not carry (cloud, radiation, precipitation phase, snow, soil moisture, pressure). Sampled at county centroid. License: CC BY 4.0.

Both sources cover the full March 11–19 window with no resolution discontinuity at storm onset.

URMA Fields (2.5 km, Polygon Mean)

Column	URMA Field	Unit	Description
gust	GUST:10m	mph	Wind gust at 10 m
wind_speed_10m	WIND:10m	mph	Sustained 10 m wind speed
wind_dir_10m	WDIR:10m	°	10 m wind direction
t2m	TMP:2m	K	2 m air temperature
d2m	DPT:2m	K	2 m dewpoint

ERA5 Fields (~31 km, County Centroid)

Column	ERA5 Name	Unit	Description
sp	surface_pressure	hPa	Surface pressure
mslma	pressure_msl	hPa	Mean sea level pressure
blh	boundary_layer_height	m	Boundary layer height
tp	precipitation	mm	Total precipitation
rain	rain	mm	Liquid rain
csnow	snowfall	cm	Snowfall
sdwe	snow_depth	m	Snow depth (water equivalent)
tcc	cloud_cover	%	Total cloud cover
lcc	cloud_cover_low	%	Low cloud cover
mcc	cloud_cover_mid	%	Medium cloud cover
hcc	cloud_cover_high	%	High cloud cover
sdsurf	shortwave_radiation	W/m ²	Downward shortwave radiation
direct_rad	direct_radiation	W/m ²	Direct solar radiation
diffuse_rad	diffuse_radiation	W/m ²	Diffuse radiation
r2	relative_humidity_2m	%	2 m relative humidity
vpd	vapour_pressure_deficit	kPa	Vapour pressure deficit
et0	et0_fao_evapotranspiration	mm	Reference evapotranspiration
soil_moist	soil_moisture_0_to_7cm	m ³ /m ³	Near-surface soil moisture

County Aggregation

- **URMA:** polygon mean over all 2.5 km grid points whose centres fall within each county boundary (point-in-polygon spatial join). All 302 counties contain at least one grid point, so every county uses a true polygon mean with no interpolation or fallback.
- **ERA5:** centroid sample; a single value extracted at each county's population-weighted centroid.

Data Sources

Source	Res.	URL
NOAA URMA (AWS S3: noaa-urma-pds)	2.5 km	https://registry.opendata.aws/noaa-rtma/
Open-Meteo ERA5	~31 km	https://open-meteo.com/en/docs/historical-weather-api
NOAA HRRR (alt)	3 km	https://registry.opendata.aws/noaa-hrrr-pds/

Infrastructure features (static, optional enrichment):

Source	URL
EIA Form 861: line miles, customers	https://www.eia.gov/electricity/data/eia861/
NLCD Tree Canopy (USGS)	https://www.mrlc.gov/data
USDA Rural-Urban Codes	https://www.ers.usda.gov/data-products/rural-urban-continuum-codes/

Notes on Resolution

URMA wind gusts are observation-corrected at 2.5 km. ERA5 fields are coarser (~31 km), which is appropriate for slowly-varying atmospheric context variables (cloud, pressure, humidity) but not suitable for the fine-scale wind fields that drive outages; those come from URMA.

URMA wind fields (`gust`, `wind_speed_10m`) are converted from their native m/s to mph ($\times 2.2369$) to match US operational convention. Wind direction remains in degrees; temperature and dewpoint (`t2m`, `d2m`) remain in Kelvin.

Important: `wind_dir_10m` is a circular variable. It ranges 0–360° where 359° and 1° are nearly the same direction (both near north) but are 358 apart numerically. Standard models will treat this incorrectly as linear. Correct encoding:

$$\text{wind_sin} = \sin\left(\frac{2\pi \times \text{wind_dir}}{360}\right), \quad \text{wind_cos} = \cos\left(\frac{2\pi \times \text{wind_dir}}{360}\right)$$

Use `wind_sin` and `wind_cos` as features instead of the raw degrees.

Note on Time Alignment

URMA timestamps are converted from UTC using a fixed −5 h (EST) offset. ERA5 data retrieved via Open-Meteo uses local America/New_York time, which is EDT (−4 h) during the event window since daylight saving began March 8, 2026. Both series are dense hourly sequences sharing the same row labels, so URMA and ERA5 values at a given labelled hour may describe physical conditions up to one hour apart. This offset is immaterial for slowly-varying ERA5 context fields such as pressure, humidity, and soil moisture, and it does not affect the URMA wind variables. It may be slightly consequential for ERA5 precipitation fields (`tp`, `rain`) during active storm passage when values change rapidly.